Understanding Differential Equations - Made Super Simple

Introduction: What is a Differential Equation?

A **differential equation** is just a special math equation that shows how something changes. Imagine you have a quantity (like temperature, population, or distance), and you want to know how fast it's changing at any moment. That's what a differential equation does—it describes that change.

Think of it like a recipe that tells you the relationship between:

- A variable (like y or temperature)
- How that variable is changing (the derivative, written as $\frac{dy}{dx}$)

For example, the equation $\frac{dy}{dx} = 2x$ means "the rate at which y changes is twice the value of x."

Real-life examples where differential equations are used:

- How fast bacteria multiply in a petri dish
- How a cup of hot tea cools down over time
- How a ball falls under gravity
- How populations grow or shrink
- How medicine spreads through your body

The most basic form is: $y = f(x)$ where y is some function that depends on x.

What's a Derivative? (Quick Refresher)

Before we go deeper, let's quickly remember what a derivative is. A **derivative** is simply the rate at which something is changing.

Symbol: $\frac{dy}{dx}$ (this is read as "dee-why dee-ex")

What it means: "How much does y change when x changes by a tiny amount?"

If you're driving a car:

- Distance is your position (y)
- Time is x
- Your speed is $\frac{dy}{dx}$ (how fast your distance changes with time)

If you see $\frac{d^2y}{dx^2}$, that means the derivative of the derivative—it shows how the rate of change itself is changing. This is called acceleration (how fast your speed is changing).

The Two Main Questions About Any Differential Equation

When you look at ANY differential equation, two questions matter:

Question 1: What is the ORDER?

Order simply means "which level of derivative is highest?"

- First-order (has $\frac{dy}{dx}$): $\frac{dy}{dx} = x + 1$
- Second-order (has $\frac{d^2y}{dx^2}$): $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$
- Third-order (has $\frac{d^3y}{dx^3}$): $\frac{d^3y}{dx^3} + \frac{dy}{dx} = 5$

How to find order: Look for the highest derivative in the equation. If the highest is $\frac{d^2y}{dx^2}$, it's second-order. Simple as that!

Question 2: What is the DEGREE?

Degree is the power (exponent) of the highest-order derivative, but ONLY if the equation is in normal mathematical form (polynomial).

Examples:

Equation	Highest Derivative	Its Power (Degree)	Order
$\left(\frac{dy}{dx}\right)^3 + \frac{dy}{dx} = x$	$\frac{dy}{dx}$	3	1
$\left(\frac{d^2y}{dx^2}\right)^2 + \frac{dy}{dx} = 0$	$\frac{d^2y}{dx^2}$	2	2
$\frac{d^3y}{dx^3} + y^2 = 5$	$\frac{d^3y}{dx^3}$	1	3

Important note: If the highest derivative has a square root or is inside a weird function, you might need to rearrange the equation first to make it polynomial.

Different Types of Differential Equations

Type 1: Ordinary Differential Equations (ODE)

An **ODE** has only ONE independent variable. For example, only x changes, and y depends on x.

Examples:

- $\frac{dy}{dx} = 3x$
- $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$

Type 2: Partial Differential Equations (PDE)

A **PDE** has more than one independent variable. For example, z might depend on both x AND y.

Examples:

- $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ (the curvy ∂ symbol means partial derivative)

Note: For Grade 12 NEB, you mainly study ODEs, not PDEs. Don't worry about PDEs too much right now.

Two Main Patterns: Linear vs Non-Linear

Linear Differential Equations

A differential equation is **LINEAR** if:

- y and its derivatives appear only to the first power
- There are no weird functions like $\sin(y)$, e^y, or y^2
- Everything can be multiplied by some function of x only

Examples of LINEAR:

- $\frac{dy}{dx} + 3y = x$
- $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$

Examples of NON-LINEAR:

- $\frac{dy}{dx} + y^2 = x$ (because of y^2)
- $\frac{dy}{dx} + \sin(y) = 0$ (because of $\sin(y)$)
- $y\frac{dy}{dx} = x$ (because y multiplies the derivative)

Linear equations are usually easier to solve, which is why they appear more in Grade 12.

Different Forms You'll See

The textbook shows different ways that differential equations can appear. Each form has its own solving method:

Form 1: Variable Separable

This is when you can get all the x terms on one side and all the y terms on the other side.

Pattern: $g(y)dy = f(x)dx$

Example: $y\,dy = x\,dx$

How to solve: Integrate both sides

- $\int y\,dy = \int x\,dx$
- $\frac{y^2}{2} = \frac{x^2}{2} + C$

Form 2: Homogeneous Differential Equations

These have a special form where you can replace $\frac{y}{x}$ with a new variable.

Pattern: $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$

How to solve: Use substitution $v = \frac{y}{x}$, then solve.

Form 3: Linear Differential Equations

These look like: $\frac{dy}{dx} + Py = Q$

Where P and Q are functions of x only.

How to solve: Use something called an "integrating factor"—a special multiplier that makes the equation solvable.

Form 4: Exact Differential Equations

These look like: $M(x, y)dx + N(x, y)dy = 0$

Where special conditions are met that make them "exact."

How to solve: There's a special technique for these (more advanced).

What is an "Initial Condition"?

An **initial condition** is like a starting point or a fact about the solution.

Example: "When $x = 3$, $y = 4$" written as $y(3) = 4$

Why does it matter?

- When you solve a differential equation, you usually get multiple possible answers (they contain constants like C)
- An initial condition lets you find the exact answer that fits your specific situation
- This is called a **particular solution**

Example:

- General solution: $y = x^2 + C$ (infinite answers, one for each value of C)
- If initial condition is $y(0) = 5$: Then $5 = 0^2 + C$, so $C = 5$
- Particular solution: $y = x^2 + 5$ (only one answer)

This is called the "Initial Value Problem" or IVP.

How to Form a Differential Equation from a Given Curve

Sometimes, you're given a general equation with constants, and you need to create the differential equation.

Steps:

1. Count how many constants you have
2. Differentiate that many times to eliminate the constants
3. What's left is your differential equation

Example 1:

- Given: $y = ax + b$ (two constants: a and b)
- Differentiate once: $\frac{dy}{dx} = a$
- Differentiate again: $\frac{d^2y}{dx^2} = 0$ ← This is your differential equation

Example 2:

- Given: $y = ax^2 + bx + c$ (three constants: a, b, c)
- Differentiate once: $\frac{dy}{dx} = 2ax + b$
- Differentiate twice: $\frac{d^2y}{dx^2} = 2a$
- Differentiate three times: $\frac{d^3y}{dx^3} = 0$ ← This is your differential equation

General Solution vs Particular Solution

Type	What is it?	Contains	Example
General Solution	All possible answers to the DE	A constant C	$y = x^2 + C$
Particular Solution	One specific answer	A specific value of C	$y = x^2 + 5$

Summary: Key Things to Remember

1. **Differential equation** = an equation showing how something changes
2. **Order** = which level of derivative? (first, second, third…)
3. **Degree** = what power is that highest derivative?
4. **Linear** = simple, no weird powers or functions of y
5. **Non-linear** = complicated, has powers or trig functions
6. **General solution** = has a constant C (many answers)
7. **Particular solution** = one specific answer (uses initial condition)
8. **Initial condition** = a starting value like $y(3) = 4$
9. **Variable separable** = you can split x and y to opposite sides

Quick Practice: Find Order and Degree

Problem 1: $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 + 6x = 0$

Solution:

- Highest derivative: $\frac{d^2y}{dx^2}$ (second-order)
- Its power: 3 (the exponent)
- **Answer: Order = 2, Degree = 3**

Problem 2: $\frac{dy}{dx} + 2y = e^x$

Solution:

- Highest derivative: $\frac{dy}{dx}$ (first-order)
- Its power: 1 (no exponent means power 1)
- **Answer: Order = 1, Degree = 1**

Difficult Words Explained Simply

Word	Means	Example
Derivative	How fast something is changing	Speed is the derivative of distance
Differential Equation	Equation with a derivative in it	$\frac{dy}{dx} = 2x$
Order	Highest level of derivative	$\frac{d^2y}{dx^2}$ has order 2
Degree	Power of the highest derivative	If it's cubed, degree = 3
Linear	Simple, first-power only	$3y + 2x = 5$
Non-linear	Complicated, has powers	$y^2 + x = 5$
Integrate	Reverse of differentiate	Opposite of finding derivative
Constant of Integration	The C in general solution	Represents all possible answers
Particular	One specific case	One answer, not infinite
Homogeneous	Has a special balanced pattern	Can be solved by substitution
Exact	Has a special property	Can be solved using exact method

Where You Go From Here

Now that you understand the basics:

- You'll learn how to actually SOLVE these equations
- Different techniques for different forms
- How to use initial conditions to find particular solutions

- Real-world applications

The key is to first identify what type of equation you have (order, degree, linear, etc.), then apply the right solving method.

Good luck with your studies! With practice, this will become much clearer.

Remember: Every expert was once a beginner. Take your time, practice the problems, and you'll master this!